date : 2026-10-05,18:26

Reference

youtube video

OpenCourseWare - Mathematics for Computer Science - 1. introduction and proofs

Handwritten note

Math for CS - 1 introduction and proofs

proofs - a method for ascertaining the truth

↓ verifying / establish

many ways of ascertaining the truth

- expectation & observation
- sampling & counter example
- judge, jury
- religion
- conviction

A math proof is verification of a proposition by a chain of logical detection from a set of axioms

Proposition

Def: a statement that is either T or F

Ex: $2+3=5 \rightarrow T$

$\forall n \in \mathbb{N}$, n^2+n+41 = prime number

"for all" ↑ "natural number" ↑ $\{0, 1, 2, \infty\}$

↓ Predicate: proposition whose T depend on the value of variable (n)

↑ quantifier

n	n^2+n+41	prime
0	41	✓
⋮	⋮	⋮
40	1681	✓

T for first 40 ⋮

39 1681 $\times (91^2)$

40

Ex: $a^4 + b^4 + c^4 = d^4$ has no positive integer solution

can be prove wrong by

$a = 95,800$
$b = 217,519$
$c = 414,560$
$d = 422,481$

correct true proposition $\Rightarrow \exists\, a,b,c,d \in \mathbb{N}^+,\ \underbrace{a^4+b^4+c^4=d^4}_{\text{Predicate}}$

"these exist" $\{1,3,\dots\}$

Ex $313\,(x^3+y^3)=z^3$ has no positive integer solution

↳ false

don't try a few cases and if it work doesn't mean its done

Ex The regions in any map can be colored in 4 colors - so that a adjcent regions have different colors

$\forall n \in \mathbb{Z},\ n \geq 3? \rightarrow n^2 \geq 4$

↑ integer $\{0,1,-1,\dots\}$ ↖ implie

→ def: An implication $p \rightarrow q$ is true if p is f or q is T

Truth table

p	q	$p \rightarrow q$
T	T	T
T	F	F


F T
F F
T

- Ex: $\forall n \in \mathbb{Z}$, $n \geq 3 \iff n^2 \geq 4$
iff

False $n = -3$ $\iff = \Leftarrow, \Rightarrow$

what is proposistion → not question

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axioms

Def: a proposition that is "assume" to be true
there is no proof just assume

Ex: If $a=b$ & $b=c$, then $a=c$

Euclidean geometry, given a line L, a point p not on L, there is only one line through p parallel to L

spherical geometry, " " " there is no line

hyperbolic geometry, " " " " There are infintely many

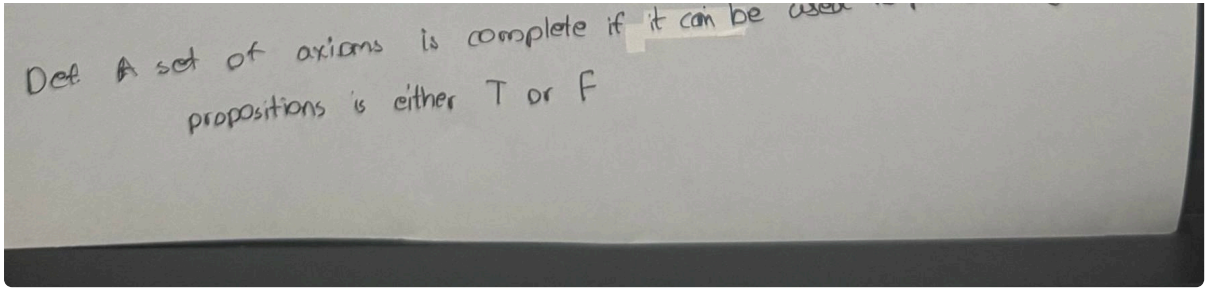
* axioms can be contradictory in different context
That is ok, state based on field

Axioms should be

1. consistent
2. coroplete

Def: A set of axioms is consistent if no proper can be proved to be both T and F

to prove every



book reading

1 Propositions

Definition : A proposition is a statement that is either true or false.

For example, both of the following statements are propositions. The first is true and the second is false.

Proposition 1.0.1 : 2 + 3 = 5.
Proposition 1.0.2 : 1 + 1 = 3

1.1 Compound Propositions

1.1.1 NOT, AND, and OR

A	NOT (A)
T	F
F	T

P	Q	P AND Q	P OR A	P XOR Q
T	T	T	T	F
T	F	F	T	T
F	T	F	T	T

P	Q	P AND Q	P OR A	P XOR Q
F	F	F	F	F

1.1.2 IMPLIES

An implication is true exactly when the if-part is false or the then-part is true.

P	Q	P IMPLIES Q	
T	**T**	**T**	(tt)
T	**F**	**F**	(tf)
F	**T**	**T**	(ft)
F	**F**	**T**	(ff)

1.1.3 IFF

P	Q	P IFF Q
T	**T**	**T**
T	**F**	**F**
F	**T**	**F**
F	**F**	**T**

1.1.4 Notation

English	Symbolic Notation
$\text{NOT}(P)$	$\neg P$ (alternatively, $\overline{P}$)
P AND Q	$P \wedge Q$
P OR Q	$P \vee Q$
P IMPLIES Q	$P \longrightarrow Q$
if P then Q	$P \longrightarrow Q$
P IFF Q	$P \longleftrightarrow Q$

1.1.5 Logically Equivalent Implications

P	Q	Proposition (P→Q)	Contrapositive (¬Q→¬P)	Converse (Q→P)	Inverse (¬P→¬Q)
T	**T**	T	T	T	T
T	**F**	F	F	T	T
F	**T**	T	T	F	F
F	**F**	T	T	T	T

an implication and its converse together are equivalent to an iff statement.

P	Q	(P↔Q)	P→Q And Q→P
T	**T**	T	T
T	**F**	F	F

P	Q	(P↔Q)	P→Q And Q→P
F	**T**	F	F
F	**F**	T	T

1.2 Propositional Logic in Computer Programs

Here is the step-by-step breakdown using standard boolean algebra / logic laws:

1. **Start with the expression:**
2. **Apply the Distributive Law:**
3. **Apply the Complement Law / Negation Law (=):**
4. **Apply the Identity Law (=):**

In summary, while it simplifies directly using the **Absorption/Redundancy Law** (=), it is proved via the **Distributive Law**, **Complement Law**, and **Identity Law**

1.3 Predicates and Quantifiers

1.3.1 Propositions with Infinitely Many Cases

Proposition 1.3.1. For every nonnegative integer, n, the value of $n^2 + n + 41$ is prime.

there are many examples of propositions that seem to be true when you check a few cases (or even many), but which turn out to be false.

> The key to remember is that you can’t check a claim about an infinite set by checking a finite set of its elements, no matter how large the finite set

it can rewrite as

$$n \in \mathbb{N}, p(n) \text{ is prime.}$$

Symbol	Name / Meaning	Explanation
$\forall$	**Universal Quantifier**	Read as "for all" or "for every".
$\in$	**Set Membership**	Read as "is a member of," "belongs to," or "is in".
$\mathbb{N}$	**Natural Numbers**	Stands for the set of *nonnegative integers*, namely $0, 1, 2, 3, \ldots$.
$p(n)$	**Function / Predicate**	An expression or value evaluated at n.

Proposition 1.3.2. *$a^4 + b^4 + c^4 = d^4$ has no solution when a, b, c, d are positive integers.*

Euler (pronounced "oiler") conjectured this proposition to be true in 1769. It was checked by humans and then by computers for many values of a, b, c, and d over the next two centuries. Ultimately the proposition was proven false in 1987 by Noam Elkies. The solution he found was $a = 95800, b = 217519, c = 414560, d = 422481$. No wonder it took 218 years to show the proposition is false!

In logical notation, Proposition 1.3.2 could be written,

$$\forall a \in \mathbb{Z}^+ \, \forall b \in \mathbb{Z}^+ \, \forall c \in \mathbb{Z}^+ \, \forall d \in \mathbb{Z}^+ . \, a^4 + b^4 + c^4 \neq d^4.$$

Here, $\mathbb{Z}^+$ is a symbol for the positive integers. Strings of $\forall$'s are usually abbreviated for easier reading, as follows:

$$\forall a, b, c, d \in \mathbb{Z}^+ . \, a^4 + b^4 + c^4 \neq d^4.$$

1.3.2 Predicates

A predicate is a proposition whose truth depends on the value of one or more variables.

use a single letter to detonate

n = "n is a perfect square"

Now P(4) is true, and P(5) is false.

1.3.3 Quantifiers

There are a couple of assertions commonly made about a predicate: *that it is sometimes true* and *that it is always true.*

Always True

For all n, $P(n)$ is true.	For all $x \in \mathbb{R}$, $x^2 \geq 0$.
$P(n)$ is true for every n.	$x^2 \geq 0$ for every $x \in \mathbb{R}$.

Sometimes True

There exists an n such that $P(n)$ is true.	There exists an $x \in \mathbb{R}$ such that $5x^2 - 7 = 0$.
$P(n)$ is true for some n.	$5x^2 - 7 = 0$ for some $x \in \mathbb{R}$.
$P(n)$ is true for at least one n.	$5x^2 - 7 = 0$ for at least one $x \in \mathbb{R}$.

an assertion that a predicate is always true, is called a *universally quantified statement.*

An assertion that a predicate is sometimes true, is called *an existentially quantified statement*

1.3.4 More Notation

Universal Quantification:

$$\forall$$

Existential Quantification:

$$\exists$$

Symbol	Name / Meaning	read as	Explanation
∀	**Universal Quantifier**	"for all" (an upside-down **A** for "**A**ll").	Indicates that P is true for all values of x in domain D
∃	**Existential Quantifier**	"there exists" (a backward **E** for "**E**xists")	Indicates that P is true for at least one value of x in domain D

The symbols $\forall$ and $\exists$ are always followed by a variable—typically with an indication of the set the variable ranges over—and then a predicate, as in the two examples above.

1.3.5 Mixing Quantifiers

1.3.5 Mixing Quantifiers

Many mathematical statements involve several quantifiers. For example, *Goldbach's Conjecture* states:

> "Every even integer greater than 2 is the sum of two primes."

Let's write this more verbosely to make the use of quantification clearer:

> For every even integer n greater than 2, there exist primes p and q such that $n = p + q$.

Let Evens be the set of even integers greater than 2, and let Primes be the set of primes. Then we can write Goldbach's Conjecture in logic notation as follows:

$$\underbrace{\forall n \in \text{Evens}.}_{\substack{\text{for every even} \\ \text{integer } n > 2}} \underbrace{\exists p \in \text{Primes}\ \exists q \in \text{Primes}.}_{\substack{\text{there exist primes} \\ p \text{ and } q \text{ such that}}} n = p + q.$$

The proposition can also be written more simply as

$$\forall n \in \text{Evens}.\ \exists p, q \in \text{Primes}.\ p + q = n.$$

1.3.6 Order of Quantifiers

- **Quantifiers of the same type can be swapped** without changing the meaning:
  - [illegible]
  - [illegible]

- **Mixed quantifiers CANNOT be swapped** without changing the meaning:
  -

How to Remember the Dependency

When reading logical expressions from left to right:

 can depend on (Each can pick its own).

 is fixed first. One single must work for all possible 's simultaneously.

1.3.7 Variable over one domain

When every variable in a logical expression belongs to the same non-empty set , repeating for every quantifier becomes redundant.

Instead, we define once as the **domain of discourse** (or simply the **domain**) and omit explicit set membership from the quantifiers:

Instead of:

Write:

1.3.8 Negating Quantifiers

1.4 Validity

A propositional formula is called *valid* when it evaluates to **T** no matter what truth values are assigned to the individual propositional variables

explanation

Two logical expressions and are **logically equivalent** () if and only if the biconditional statement is **valid**.

In the text, the Distributive Law states:

To prove this law using validity, we check whether the biconditional formula is valid:

use True table

The same idea extends to predicate formulas, but to be valid, a formula now must evaluate to true no matter what values its variables may take over any unspecified domain, and no matter what interpretation a predicate variable may be given. For example, we already observed that the rule for negating a quantifier is captured by the valid assertion (1.5).

Another useful example of a valid assertion is

$$\exists x \forall y.\ P(x, y) \text{ IMPLIES } \forall y \exists x.\ P(x, y). \tag{1.7}$$

Here's an explanation why this is valid:

Let D be the domain for the variables and P_0 be some binary predicate[4] on D. We need to show that if

$$\exists x \in D \; \forall y \in D. \; P_0(x, y) \tag{1.8}$$

holds under this interpretation, then so does

$$\forall y \in D \; \exists x \in D. \; P_0(x, y). \tag{1.9}$$

So suppose (1.8) is true. Then by definition of $\exists$, this means that some element $d_0 \in D$ has the property that

$$\forall y \in D. \, P_0(d_0, y).$$

By definition of $\forall$, this means that

$$P_0(d_0, d)$$

is true for all $d \in D$. So given any $d \in D$, there is an element in D, namely, d_0, such that $P_0(d_0, d)$ is true. But that's exactly what (1.9) means, so we've proved that (1.9) holds under this interpretation, as required.

1.5 Satisfiability

A proposition is **satisfiable** *if some setting of the variables makes the proposition true*

The general problem of deciding whether a proposition is satisfiable is called **SAT**.

approaches to **SAT**

1. to construct a truth table and check whether or not a T ever appears.